



13-CONTRACT-FREEZE-AND-ADR.md — Stopping drift across 30 phases



The single biggest cause of failure in a 30-phase agent-built system is **contract drift**: phase 22 "improves" a struct that phases 09, 14 and 27 depend on. This file makes every shared type a locked artefact with a change ritual.

1. What counts as a frozen contract

Kind	Location	Frozen from
Public Rust types crossing a crate boundary	<code>crates/*/src/contract/**</code>	Phase that introduces them
SQLite schema (tables, columns, indexes)	<code>migrations/*.sql</code>	Phase 01
Edit-recipe JSON v1	<code>schemas/recipe.v1.json</code>	Phase 14
Tauri IPC commands and events	<code>crates/aura-app/src/ipc/**</code>	Phase 01
Model I/O signatures	<code>models/<name>/signature.json</code>	Phase 03
Cloud task prompts + response schemas	<code>schemas/cloud/*.json</code>	Phase 04
Error codes	<code>crates/aura-core/errors.toml</code>	Phase 01
Telemetry event names + fields	<code>schemas/telemetry.json</code>	Phase 01
Export sidecar format	<code>schemas/sidecar.v1.json</code>	Phase 30

Anything else is free to refactor without ceremony. Keeping the frozen set *small and explicit* is what makes the ritual survivable.

2. `contracts.lock`

A generated file at the repo root holding a hash per contract artefact:

```
# contracts.lock – GENERATED. Do not edit by hand.
schema_version = 1

["crates/aura-core/src/contract/photo.rs"]
sha256 = "9f2c..."
frozen_in = "PHASE-01"
adr = []

["schemas/recipe.v1.json"]
sha256 = "41ab..."
frozen_in = "PHASE-14"
adr = ["ADR-0007"]
```

CI gate `contract-freeze`: recompute hashes. If any differ and the PR does not contain a new `docs/adr/ADR-XXXX-*.md` referencing that exact path, the build fails with:

```
FROZEN CONTRACT CHANGED: schemas/recipe.v1.json
  expected 41ab... found 88de...
  No ADR in this PR references this path. Write one or revert.
```

3. The ADR template (copy verbatim)

```
# ADR-0007 – Add per-mask opacity to recipe v1

- Status: Proposed | Accepted | Superseded by ADR-XXXX
- Date: 2026-08-09
- Deciders: CTO (required), TLC, + owner of each affected contract
- Phase raised in: PHASE-18
- Contracts touched: schemas/recipe.v1.json, crates/aura-rend
```

```
er/src/contract/mask.rs
```

Context

What forced this. Include the failing case, not just a preference.

Options considered

1. Do nothing – consequences.
2. Additive change – consequences.
3. Breaking change – consequences.

Decision

Exactly what changes, with the before/after type written out.

Compatibility

- Additive / Breaking:
- Migration for existing catalogs:
- Recipe version bump required? y/n
- Golden manifest regeneration required? y/n
- Model repackaging required? y/n

Blast radius

List every phase, crate, test and fixture that must change. Grepping evidence required.

Rollback

How we undo this if it lands badly.

Consequences

What becomes easier, what becomes harder.

4. Change classes and what each requires

Class	Example	Required
Additive-optional	New optional field with a default	ADR (short form), lock update, no version bump
Additive-required	New required field	ADR + migration + recipe minor bump + fixture update
Semantic	Same field, new meaning or range	ADR + major bump; renaming the field is preferred over reinterpreting it
Breaking	Remove/rename/retype a field	ADR + major bump + forward migration + read-old-write-new shim for 2 releases

Golden rule: never reinterpret an existing field. Add a new one and deprecate the old. Reinterpretation is invisible to the compiler and to Claude Code, which makes it the most dangerous change in the codebase.

5. Compatibility rules by artefact

- **SQLite** — forward-only numbered migrations, each idempotent and wrapped in a transaction. Every migration ships with a downgrade note (even if the note is "restore from the pre-migration backup we take automatically"). The app refuses to open a catalog with a newer `user_version` ([AURA-DB-3002](#)).
- **Recipe JSON** — readers must accept unknown fields and preserve them on write (lossless round-trip test required). A recipe written by a newer app opened in an older app must render *correctly or refuse*, never *wrongly*.
- **IPC** — commands are versioned by name (`develop.render.v2`), never by silently changing arguments. Both versions coexist for one release.
- **Models** — changing input size, normalisation, or output layout is always a new model *name*, not a new version of the same name.

6. Ownership matrix

Contract	Owner	Mandatory reviewer
Catalog schema, job graph types	SRC	TLC, CTO
Recipe JSON, colour pipeline types	COL	SRG, QAIQ
Model signatures, package manifest	MLOPS	MLL, SEC

Contract	Owner	Mandatory reviewer
Cloud task schemas, cost governor	AGT	SEC, CTO
IPC surface, event names	SFE	TLC
Error registry	TLC	QAL, DOC
Telemetry schema	DEVOPS	SEC, PM

7. Deprecation lifecycle

1. **Mark** — `#[deprecated(since = "0.9.0", note = "use X; ADR-0011")]`, schema field annotated `"deprecated": true`.
2. **Warn** — one release with both paths live and a telemetry counter on the old path.
3. **Migrate** — automatic on catalog open, logged in the ledger.
4. **Remove** — earliest two minor releases after marking, in a dedicated PR that touches nothing else.

Nothing is removed while telemetry shows the old path still in use above 0.1 % of runs.

8. Test plan

- **Lock integrity** — `cargo xtask contracts --check` reproduces `contracts.lock` exactly.
- **Round-trip** — recipe v1 fixtures with unknown future fields survive read → write byte-identically.
- **Migration chain** — a catalog created at every past schema version migrates to head and passes `PRAGMA integrity_check`.
- **Old-app guard** — opening a newer catalog raises `AURA-DB-3002` and never writes.
- **ADR linter** — every ADR has all required sections and a resolvable `Contracts touched` path list.

9. Acceptance criteria

- ☐ `contracts.lock` exists and is generated by `cargo xtask contracts`.
- ☐ `contract-freeze` CI gate blocks unaccompanied contract changes.
- ☐ `docs/adr/` contains ADR-0001 recording the freeze policy itself.
- ☐ Round-trip and migration-chain tests are green.
- ☐ Every frozen file lists its owner in `CODEOWNERS`.

10. Claude Code prompt

Act as `CTO`. Implement the `cargo xtask contracts` generator and the `contract-freeze` CI gate exactly as described in `13-CONTRACT-FREEZE-AND-ADR.md`, then generate the initial `contracts.lock` from the current repo and write `docs/adr/ADR-0001-contract-freeze-policy.md` using the template verbatim. From now on, before editing any path listed in `contracts.lock`, stop and produce the ADR first — including a real grep-based blast radius — and wait for my approval. Never update `contracts.lock` in the same commit as unrelated code.